

STEP 1: Install Libraries

```
%pip install PyPDF2 nltk scikit-learn pandas plotly nbformat
```

✓ 7.4s Python

STEP 2: Import Libraries

```
import PyPDF2
import re
import string
import pandas as pd
import nltk
import plotly.express as px

from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import OneHotEncoder
```

✓ 7.0s Python

STEP 3: Download NLTK Data

```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
```

✓ 0.7s Python

STEP 4: Download the PDF and put it in the same folder

STEP 5: Read PDF

```
pdf_path = "book.pdf"

text = ""

with open(pdf_path, "rb") as file:
    reader = PyPDF2.PdfReader(file)

    total_pages = len(reader.pages)

    pages_to_read = min(100, total_pages)

    for page_num in range(pages_to_read):
        page = reader.pages[page_num]
        text += page.extract_text()

print("Total Pages in PDF:", total_pages)
print("Pages Extracted:", pages_to_read)
```

✓ 6.1s Python

Total Pages in PDF: 212
Pages Extracted: 100

STEP 6: Show Sample Text

```
print(text[:1000])
```

✓ 0.0s Python

THE AGENTIC AI
HANDBOOK
Concepts, Design Patterns, and Future
Directions

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ISBN: 9798265909534
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STEP 7: LOWERCASE

```
lower_text = text.lower()  
  
print(lower_text[:500])
```

✓ 0.0s Python

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STEP 8: REMOVE NUMBERS (Regex)

```
no_numbers = re.sub(r'\d+', '', lower_text)

print(no_numbers[:500])
```

✓ 0.0s Python

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STEP 9: REMOVE SPECIAL SYMBOLS

```
no_special = re.sub(r'[^a-zA-Z\s]', '', no_numbers)

print(no_special[:500])
```

✓ 0.0s Python

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STEP 10: REMOVE EXTRA SPACES

```
clean_text = re.sub(r'\s+', ' ', no_special).strip()

print(clean_text[:500])
```

✓ 0.0s Python

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STEP 11: REMOVE PUNCTUATION

```
clean_text = clean_text.translate(
    str.maketrans('', '', string.punctuation)
)

print(clean_text[:500])
```

✓ 0.0s Python

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STEP 12: TOKENIZATION

```
tokens = word_tokenize(clean_text)

print(tokens[:50])
```

✓ 0.3s Python

['the', 'agentic', 'ai', 'handbook', 'concepts', 'design', 'patterns', 'and', 'future', 'directions', 'kaushik', 'bar', 'isbn',

STEP 13: STOP WORD REMOVAL

```
stop_words = set(stopwords.words('english'))

valid_words = [
    word for word in tokens
    if word not in stop_words
]

stopword_count = len(tokens) - len(valid_words)

print("Stop Words Count:", stopword_count)
print("Valid Words Count:", len(valid_words))
```

✓ 0.0s Python

Stop Words Count: 3633
Valid Words Count: 9522

STEP 14: STEMMING

```
stemmer = PorterStemmer()

stemmed_words = [
    stemmer.stem(word)
    for word in valid_words
]

print(stemmed_words[:50])
```

✓ 0.5s Python

['agent', 'ai', 'handbook', 'concept', 'design', 'pattern', 'futur', 'direct', 'kaushik', 'bar', 'isbn', 'copyright', 'kaushik'

STEP 15: LEMMATIZATION

```
lemmatizer = WordNetLemmatizer()

lemmatized_words = [
    lemmatizer.lemmatize(word)
    for word in valid_words
]

print(lemmatized_words[:50])
```

✓ 8.4s Python

['agentic', 'ai', 'handbook', 'concept', 'design', 'pattern', 'future', 'direction', 'kaushik', 'bar', 'isbn', 'copyright', 'ka'

STEP 16: ONE HOT ENCODING

```
sample_words = list(set(lemmatized_words[:20]))

encoder = OneHotEncoder(sparse_output=False)

encoded = encoder.fit_transform(
    pd.DataFrame(sample_words)
)

onehot_df = pd.DataFrame(
    encoded,
    columns=encoder.categories_[0]
)

print(onehot_df)
```

✓ 0.1s Python

	agentic	ai	bar	concept	copyright	design	direction	disclaimer	\
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
1	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	
6	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	
7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
8	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	
9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
10	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	
11	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	

STEP 17: TF-IDF

```
final_text = " ".join(lemmatized_words)

vectorizer = TfidfVectorizer(max_features=20)

tfidf_matrix = vectorizer.fit_transform([final_text])

feature_names = vectorizer.get_feature_names_out()

tfidf_scores = tfidf_matrix.toarray()[0]

tfidf_df = pd.DataFrame({
    "Word": feature_names,
    "TF-IDF Score": tfidf_scores
})

print(tfidf_df)
```

✓ 0.0s Python

	Word	TF-IDF Score
0	action	0.159097
1	agent	0.301149
2	agentic	0.462140
3	ai	0.500021
4	based	0.113641
5	concept	0.132581
6	design	0.115535
7	environment	0.138263
8	future	0.134475
9	goal	0.185614

STEP 18: GRAPHS

SCATTER PLOT

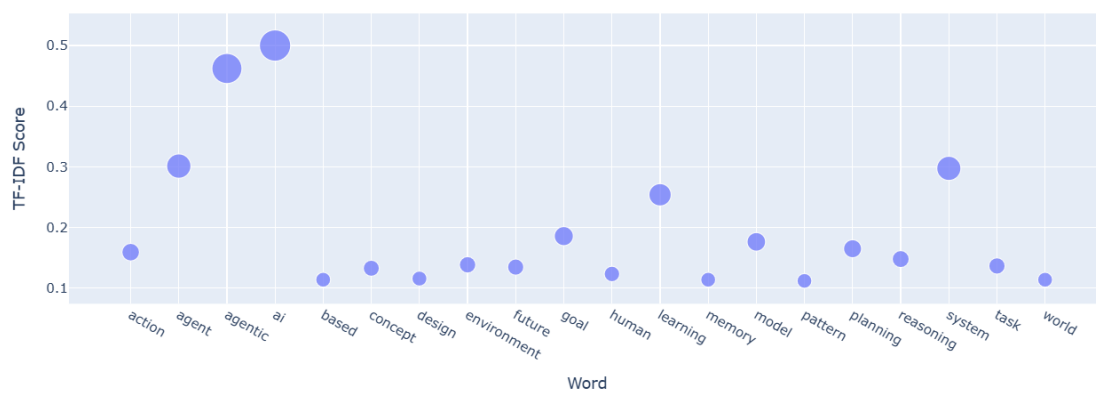
```
fig = px.scatter(  
    tfidf_df,  
    x="Word",  
    y="TF-IDF Score",  
    title="TF-IDF Scatter Plot",  
    size="TF-IDF Score"  
)
```

```
fig.show()
```

✓ 2.5s

Python

TF-IDF Scatter Plot



BAR GRAPH

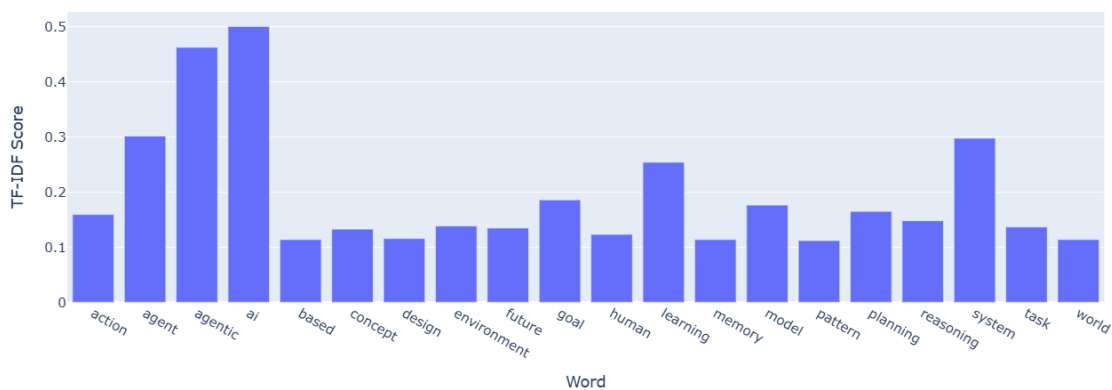
```
fig = px.bar(  
    tfidf_df,  
    x="Word",  
    y="TF-IDF Score",  
    title="Top TF-IDF Words"  
)
```

```
fig.show()
```

✓ 0.2s

Python

Top TF-IDF Words



LINE GRAPH

```
fig = px.line(  
    tfidf_df,  
    x="Word",  
    y="TF-IDF Score",  
    title="TF-IDF Line Graph"  
)
```

```
fig.show()
```

✓ 0.2s

Python

